Bindcraft Ewok Setup

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1. Prerequisites

1.1 Install VS Code

Download and install VS Code from the official download page:

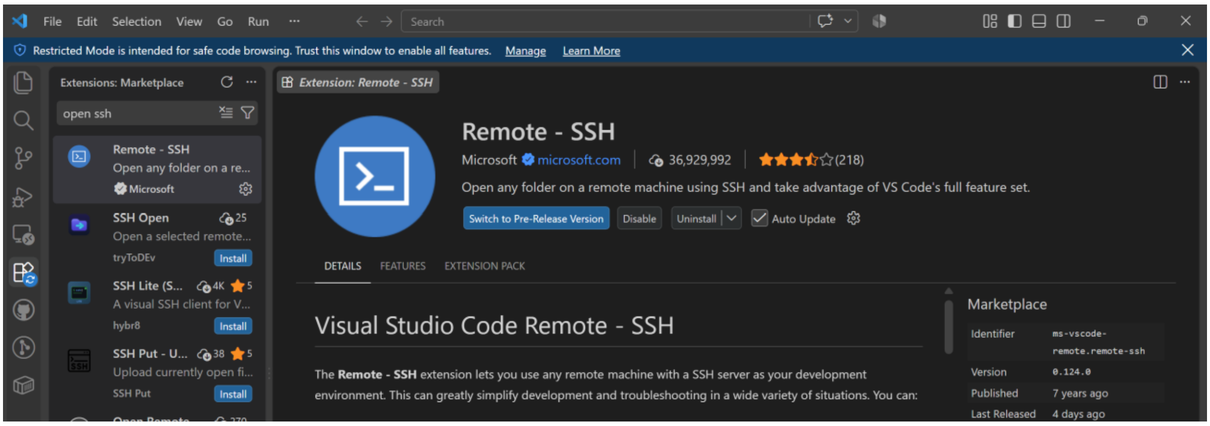
https://code.visualstudio.com/download?_exp_download=fb315fc982

1.2 Github Account

Create a GitHub account at https://github.com/ if you do not already have one.

1.3 Remote- SSH extension

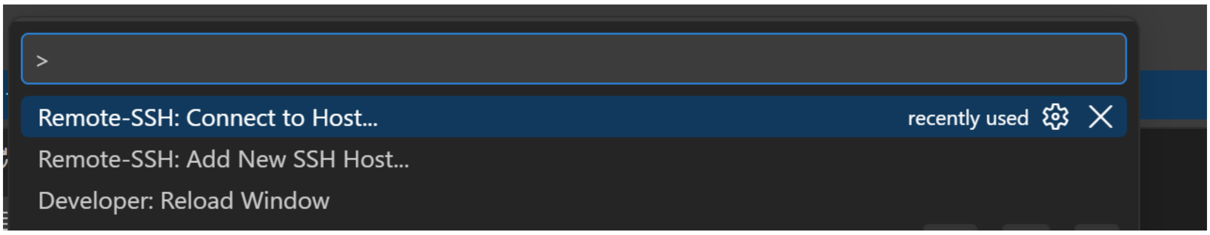
Install the **Remote – SSH** extension in VS Code. This extension enables you to connect to an Ewok machine from your local computer and access its computational resources directly through VS Code.

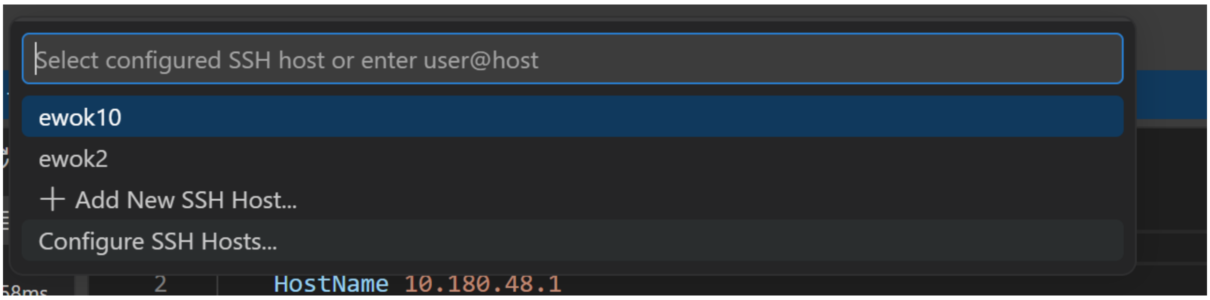


After installation, open the **Command Palette** and run **Developer: Reload Window** to ensure that the extension loads correctly.

2. Connect to the Ewok Machine

In VS Code, open the **Command Palette** and select **Remote-SSH: Connect to Host…**, followed by **Configure SSH Hosts…**.





Add an entry for your Ewok machine using the following configuration format:

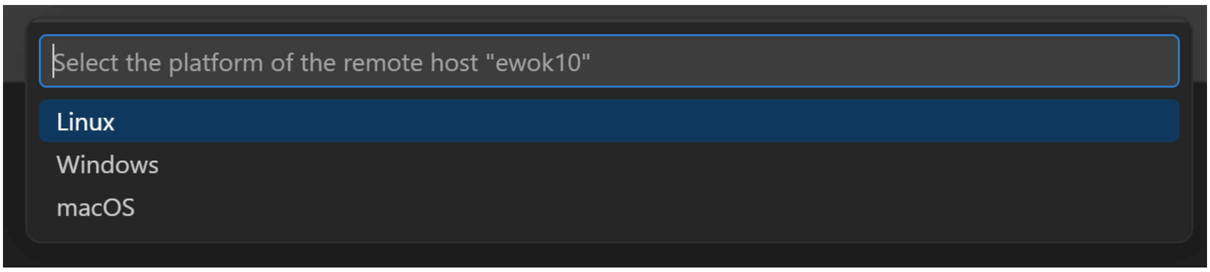
```
Host <ewok_machine_number>
  HostName <your_ewok_machine_ip_address>
  User <your_ewok_username>
  Port <your_ewok_port>
```

Example configuration:

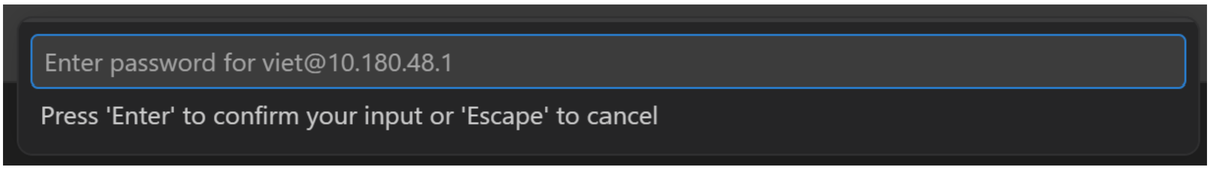
```
Host ewok2
  HostName 11.181.47.0
  User viet
  Port 11111
```

Note: Contact [Zhen He]() or [Aaron Haris]() for setting up and access your Ewok account.

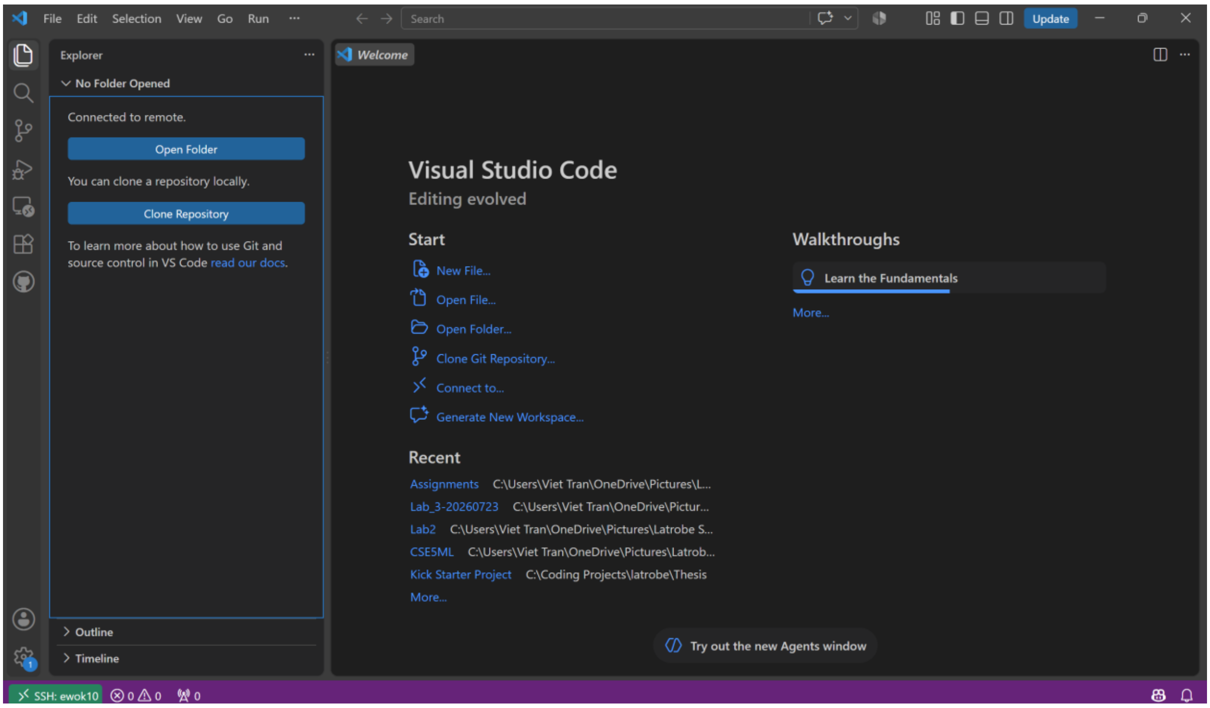
If VS Code prompts you to select the remote host's operating system, choose **Linux**



Enter your Ewok password to complete the connection.

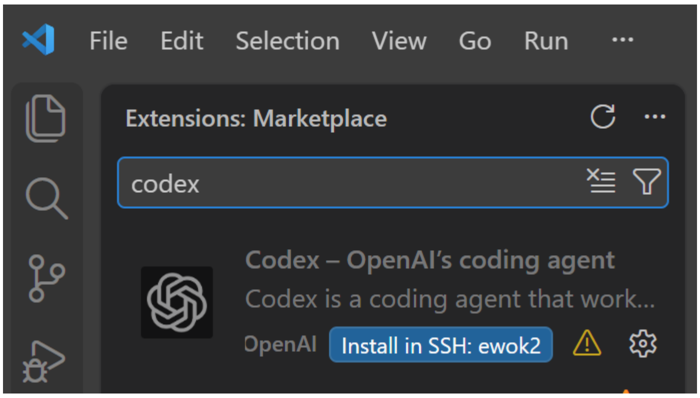


Once the connection is established, the Ewok host name should appear in the bottom-left corner of the VS Code window.



Optional: Install Codex

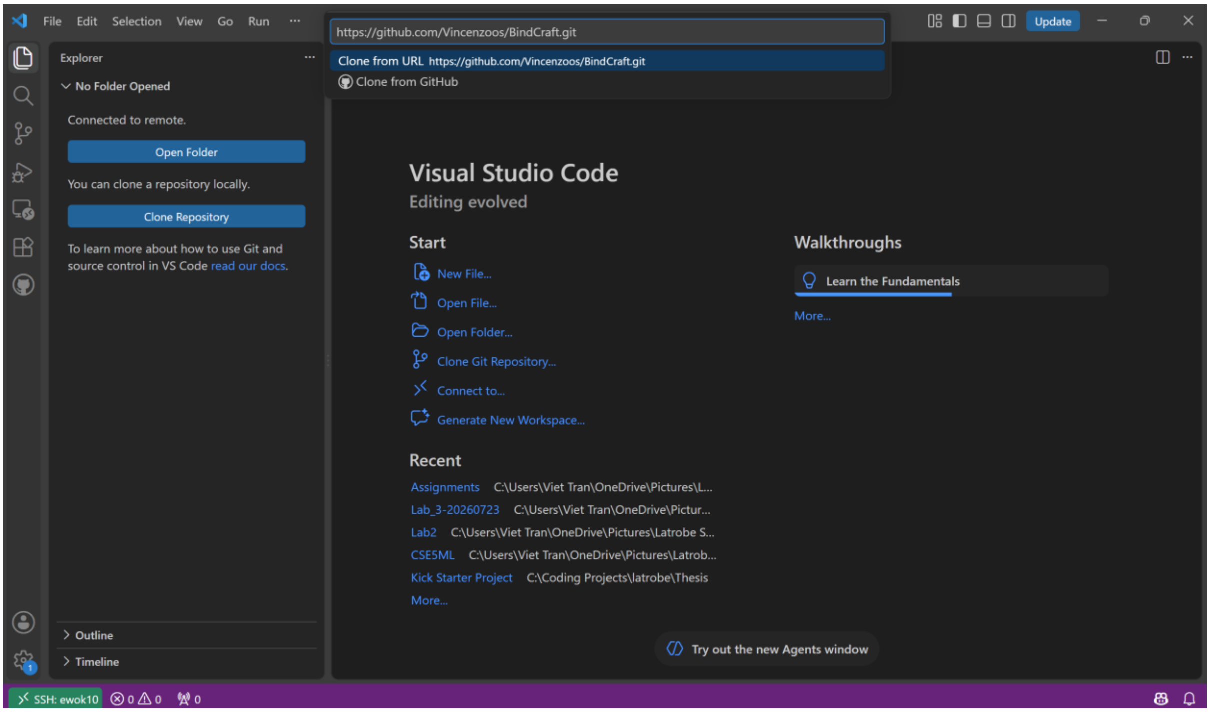
If you have a ChatGPT account, you can optionally install the Codex extension on the Ewok host for AI-assisted work within the project.



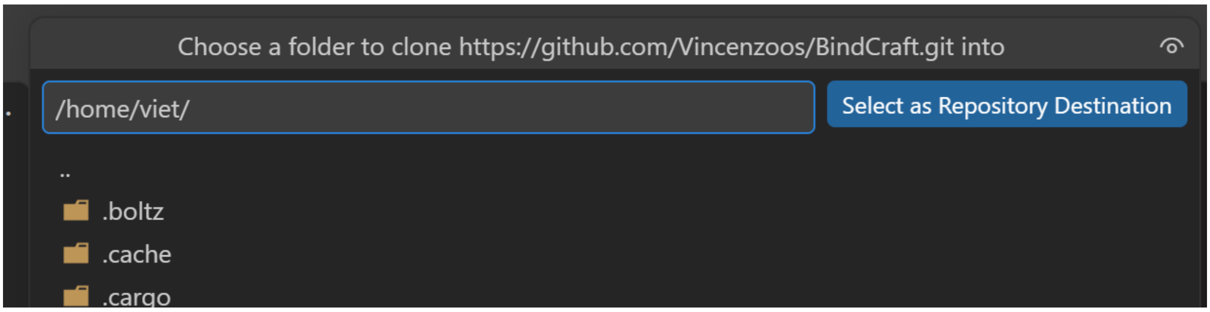
3. Clone the BindCraft Repository

After connecting to Ewok, clone the BindCraft repository from GitHub onto the remote machine. Github repository: https://github.com/Vincenzoos/BindCraft.git

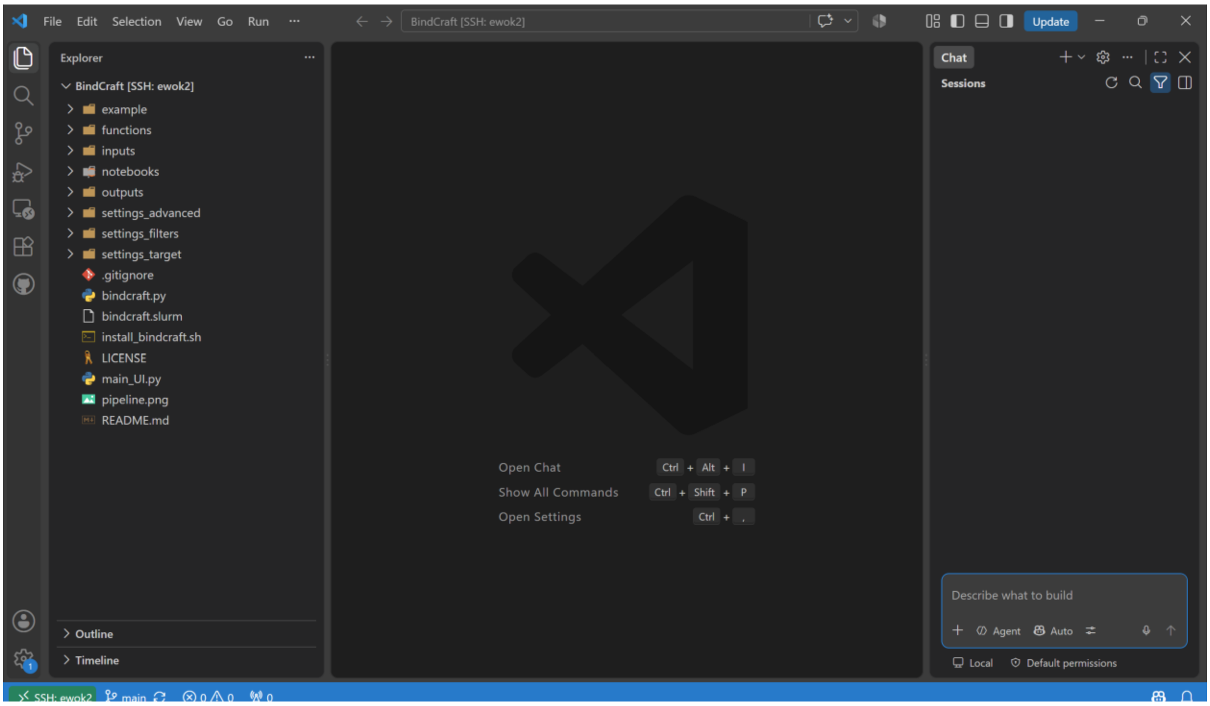
In the left sidebar, open Explorer, select Clone Repository, and enter the repository URL when prompted.



/home or /storage are two suitable location to store the project. /storage is preferred if additional disk space needed.



Once cloning is complete, the Explorer panel should display the complete BindCraft file structure.



4. Set Up the BindCraft Environment

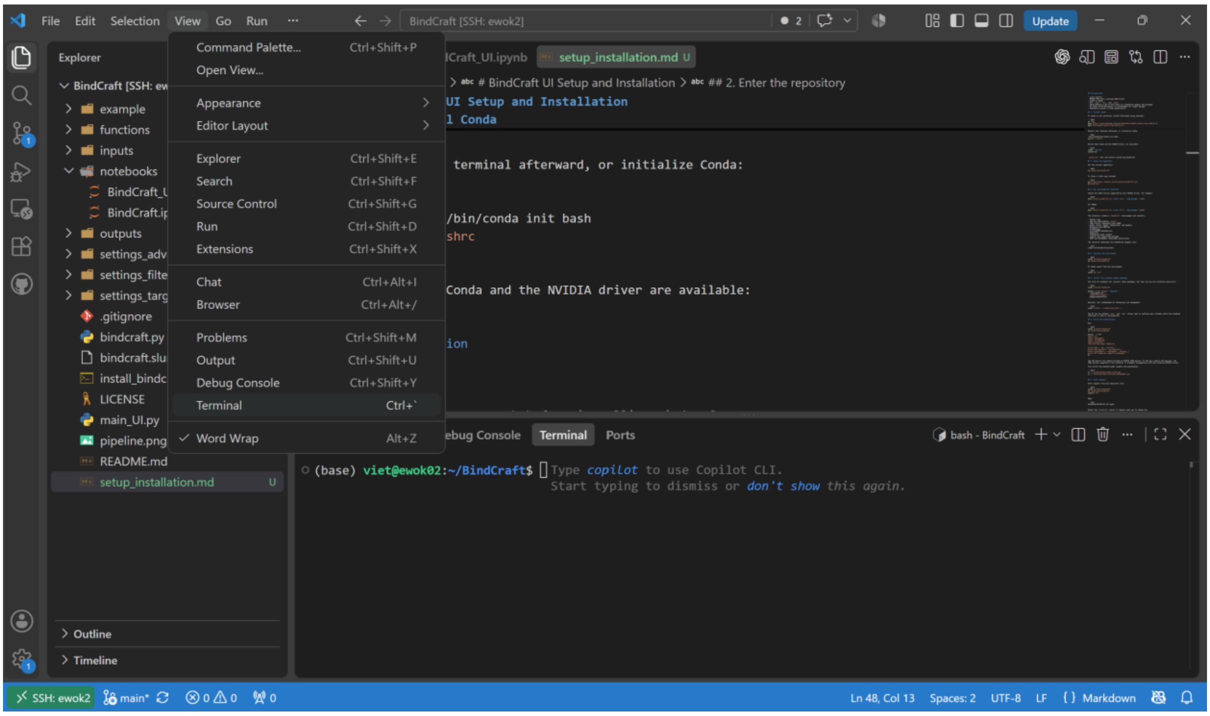
Before running the BindCraft UI notebook, install the BindCraft dependencies and the notebook UI dependencies.

4.1 Install Conda

If Conda is not installed on the Ewok machine, install Miniconda using the terminal:

```
cd /tmp
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
bash Miniconda3-latest-Linux-x86_64.sh
```

To access the terminal in VS Code, navigate to the bottom panel and expand it, or select View > Terminal.



Run the following commands to verify that Conda and the NVIDIA driver are available:

```
conda --version
nvidia-smi
```

Expected result:

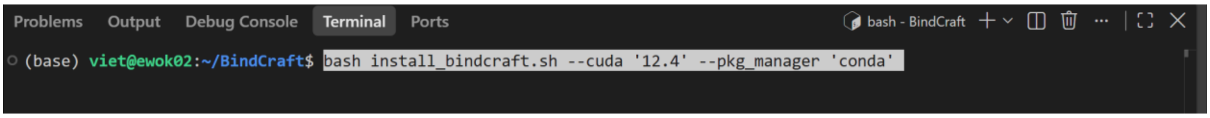
```
(base) viet@ewok02:~/BindCraft$ conda --version
nvidia-smi
conda 26.5.3
Wed Aug 26 10:05:18 2026
+---------------------------------------------------------------------------------------+
| NVIDIA-SMI 535.274.02             Driver Version: 535.274.02   CUDA Version: 12.2     |
|-----------------------------------------+----------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |         Memory-Usage | GPU-Util  Compute M. |
|                                         |                      |               MIG M. |
|=========================================+======================+======================|
|   0  NVIDIA GeForce RTX 4080 ...    Off | 00000000:0B:00.0 Off |                  N/A |
|  0%   47C    P8               9W / 320W |      1MiB / 16376MiB |      0%      Default |
|                                         |                      |                  N/A |
+-----------------------------------------+----------------------+----------------------+
|   1  NVIDIA GeForce RTX 4080 ...    Off | 00000000:41:00.0 Off |                  N/A |
|  0%   41C    P8              11W / 320W |      1MiB / 16376MiB |      0%      Default |
|                                         |                      |                  N/A |
+-----------------------------------------+----------------------+----------------------+

+---------------------------------------------------------------------------------------+
| Processes:                                                                            |
|  GPU   GI   CI        PID   Type   Process name                            GPU Memory |
|        ID   ID                                                             Usage      |
|=======================================================================================|
|  No running processes found                                                           |
+---------------------------------------------------------------------------------------+
```

4.2 Run the BindCraft Installer

Specify a CUDA version supported by the NVIDIA driver on the Ewok machine. For example:

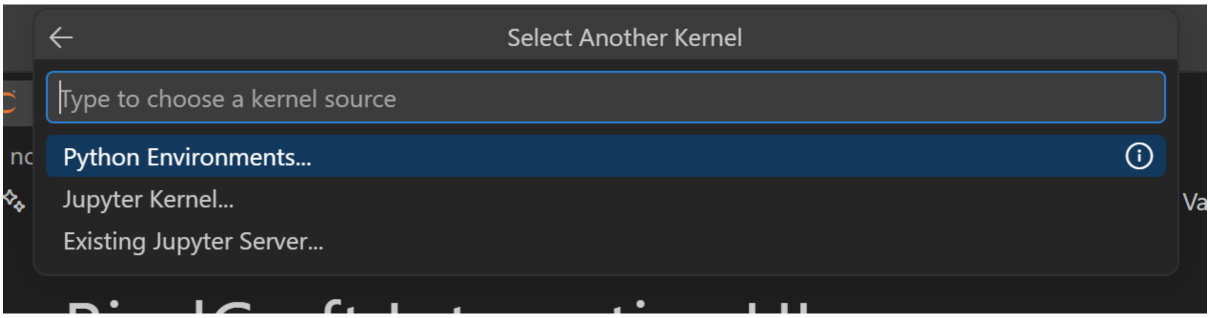
```
bash install_bindcraft.sh --cuda '12.4' --pkg_manager 'conda'
```

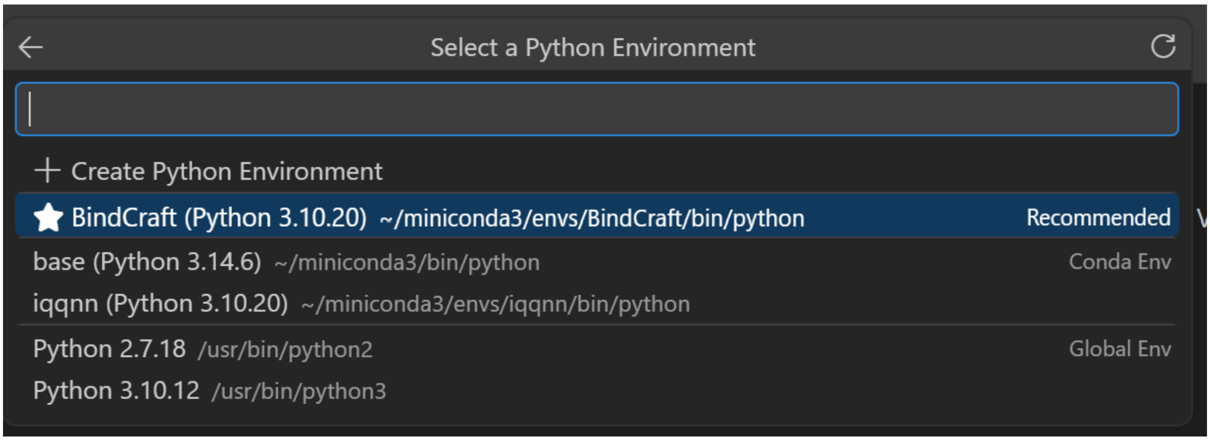


4.3 Run the BindCraft UI Notebook

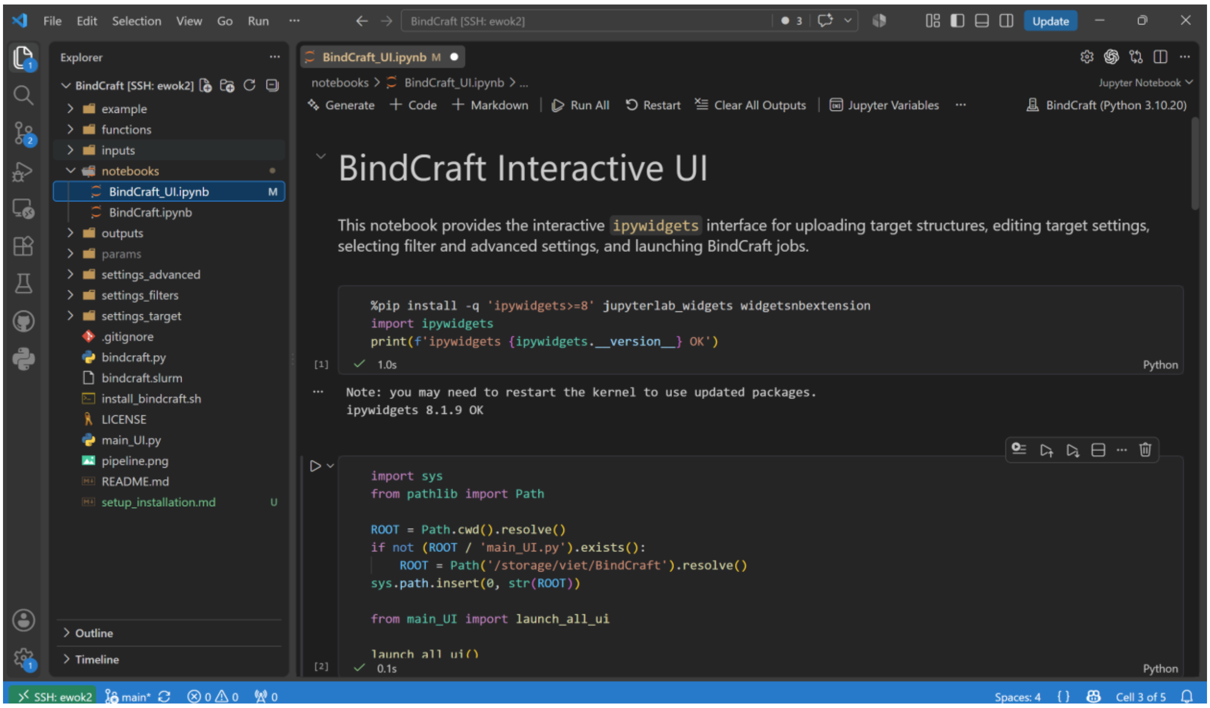
Open the Bindcraft_UI.ipynb notebook in the notebooks folder.

Select the bindcraft Conda environment as the notebook kernel:





Then select **Run All** cells for dependencies installation and UI rendering.



If the UI does not appear, you may need to reload the notebook to ensure that the ipywidgets dependency is loaded correctly.